

**IsDB-BISEW: Islamic Development Bank Bangladesh
Islamic Solidarity Educational Wakf**

Case Study ON

Design and Implementation of a Food Delivery
Management System Using SQL



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Microsoft SQL Server*

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INTRODUCTION

FDMS is a food delivery service platform dedicated to connecting customers, vendors, and delivery riders in a seamless way. To improve operational efficiency and support scalability, FDMS has implemented a comprehensive database-driven system. This system provides real-time order tracking, ensures data consistency, and facilitates informed decision-making

Objectives

- **Efficiency:** Automate and streamline order processing and delivery operations.
- **Data Accuracy:** Ensure reliable, validated data entry with minimal redundancy.
- **Customer Experience:** Provide smooth interaction between customers, vendors, and riders.
- **Sales & Payment Management:** Accurately track orders, payments, and rider commissions.
- **Analytics & Reporting:** Provide insights into vendor sales, order volumes, and rider performance.

Summary

FDMS is a digital platform designed to manage food delivery services by integrating technology into restaurant and courier operations. The Food Delivery Management System helps restaurants, customers, and delivery riders streamline orders, track payments, improve efficiency, and maximize profitability.



Database Design

➤ Entities:

Customers, Riders, Vendors, Categories, Menu Items, Orders, Order Items, Payments, Rider Payments, Reviews.

Relationships:

- One customer can place multiple orders.
- One order belongs to one vendor and is assigned to one rider.
- -One order can include multiple menu items.
 - Payments are linked to orders.

- Riders receive payment only for delivered orders.

***Normalization:** The design reduces redundancy using primary keys, foreign keys, unique constraints, and check constraints.

Challenges and Solutions

- 1. Data Accuracy: Prevented with validation checks and constraints.
- 2. Order Tracking: Automated triggers update order totals and delivery status.
- 3. User Adoption: Simplified stored procedures for easy use and adoption

Reports and Analytics

FDMS generates reports for: - Financials: Vendor revenues, rider commissions, and customer bills. - Orders: Pending, delivered, and cancelled orders. - Customer Trends: High-value and repeat customers. - Vendor Performance: Sales breakdown by category and restaurant

USPs and Benefits

- Comprehensive Management across customers, vendors, riders, orders, and payments. - Real-Time Updates using triggers. – User-Friendly for fast CRUD operations. – Scalable for multiple vendors and large transactions. – Reliable with strong constraints and referential integrity. Business Requirements 1. Customer Order Management.
- 2. Rider Assignment with payment validation.
- 3. Vendor Operations and menu management.
- 4. Sales & Payment Tracking. 5. Review Collection. 6. Reporting and Analytics.

Case Study Walkthrough

1. Initial Setup – Vendors, riders, customers, and categories created.
2. Order Placement – Orders linked with menu items.
3. Payment and Rider Commission – Recorded through Payments and RiderPayments tables.
4. Review Submission – Ratings stored in Reviews.
5. Reports – Views and UDFs generate insights. Conclusion The Food Delivery Management System provides a robust and scalable solution for food delivery

businesses. Using DDL, DML, constraints, procedures, triggers, and functions, FDMS ensures accuracy and efficiency. It can be extended to real-world platforms like Foodpanda, Uber Eats, or DoorDash.

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Conclusion

The Fisheries Farm Management System for eFarm.com provides a comprehensive solution for managing fish farming operations. By leveraging data analytics, real-time monitoring, and automation, it enables fisheries managers to optimize resources, reduce waste, and boost productivity.

**Our greatest glory is
not in never failing,
but in rising every
time we fail.**

- Confucius

